

A comparison of analytical models for crack-width prediction in RC ties subject to direct tension and restrained shrinkage

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ABSTRACT

With growing relevance to the serviceability design of important infrastructure projects such as nuclear power plants, bridges, tall buildings, and water retaining structures, accurate prediction of crack width is gaining importance. However, the prediction of crack width in reinforced concrete (RC) members is rendered difficult due to material uncertainties within the temporal and spatial domain. Empirical models are usually employed to predict the crack width in RC members; however, they are limited within the bounds of the regression obtained through experimental studies. This study comprehensively compares ten analytical models to predict crack widths in RC ties subjected to direct tension and restrained shrinkage. The analytical models are grouped into two categories: (i) assuming no-slip condition and (ii) assuming bond-slip between the steel and concrete. A database of experiments carried out on RC ties is first collated. The analytical models are then compared with the existing experimental database, and their performance is evaluated. Further, the limitations in each model are thoroughly examined and demonstrated with the help of a detailed parametric study. It is seen that most analytical models do not adequately include the effect of reinforcement yielding, especially when low reinforcement percentages are adopted in the RC ties. It is seen that the Pure Shear Model (PSM) and Tension Chord Model (TCM) show a MAPE value of 0.33 and 0.32, respectively, when compared with the experimental database. The use of a more refined Modified Tension Chord Model (MTCM) does not necessarily improve the crack-width prediction. This study aims to identify the anomalies and limitations in the analytical models, which can then be refined further. Improving these analytical models can enable their integration into design codes.

Keywords: Crack Width, PSM, TCM, MTCM, Analytical, RC Ties, and Serviceability.

INTRODUCTION:

The cracking of reinforced concrete members poses a significant challenge to the serviceability, safety and the aesthetics of any structure. What exacerbates the challenge is the accurate prediction of the crack widths in the design stage to avoid catastrophic results later. Design codes around the world are based on empirical or semi-empirical equations and show a poor accordance with the experimental results [1]. This study dives into the existing analytical models to predict the crack width for members subjected to direct tension and restrained shrinkage. The analytical models differ from one another in their bond-slip assumptions, variations of stresses in steel and concrete, the philosophy used in defining the crack width, and other underlying assumptions which are discussed in the sections to

come. The analytical models evaluated in this study date back to 1977 [2] and are compared with recent works to see how the assumptions have evolved over time and to find out whether these sophisticated assumptions and open-form solutions show a significant improvement of the crack width over the simpler models. After an extensive literature review, this study lays down the assumption that govern each model and evaluate the theoretical crack-width predicted by these models against the experimental crack widths obtained of members in direct tension and restrained shrinkage. A Sobol's sensitivity analysis [3] using the Latin Hypercube Sampling technique is used to rank the parameters affecting the crack-width. Further, the shortcomings in each model are outlined that paves the way forward to deriving a more robust analytical model to predict the crack-width and thus make an improvement in the existing design codes.

DIRECT TENSION MODELS:

Governing Philosophies:

1. Based on the deformation in Steel:

a. Leonhardt [2]:

Based on Goto's tests [4] and findings on internal cracking, Leonhardt divided a cracked segment into two parts: Lost bond region and Transfer length region. The lost bond region is defined as the immediate region around the crack where the bond between the steel and concrete is lost, thus, there is no bond-stress here, whereas the transfer length region denotes the region between the cracks where the bond stress is active. Thus, the crack-width is defined as the change in length of steel over these two regions, i.e.,

$$w = l_o \varepsilon_s'' + l_{tr} \varepsilon_m \quad (1)$$

where ε_s'' is the steel strain at the crack, and ε_m is the mean steel strain in the transfer length. ε_m is found by subtracting the tension stiffening by concrete from the ε_s'' . The model uses the following assumptions:

I. $l_o = \frac{\sigma_{sr}}{45} \phi$. It is mentioned that this value needs to be checked by further tests, thus, presents a scope for improvement.

II. $l_{tr} = \frac{f_{ct} A_{ct}}{\tau_{b,av} \Sigma u} + K_1(c, a)$, where the first term is found by using the equilibrium condition and the second term is introduced to include the effect of cover and bar spacing. This equation is reduced to: $l_{tr} = K_1(c, a) + K_2 K_3 \frac{\phi}{\rho}$, where $K_2 = \frac{f_{ct}}{\tau_{b,av}}$ and $\frac{A_{ct}}{\Sigma u} = K_3 \frac{\phi}{\rho}$. These constants are found by having values based on practical consideration depending on the bar type (ribbed or smooth) and the loading case (pure tension or pure bending) respectively.

2. Based on Solving the 2nd order differential equation for slip:

The analytical models follow the same philosophy for finding the crack widths, i.e., they first derive the Second Order Differential Equation for slip using the equilibrium criterion. Then these models solve the SODE for a particular bond-slip relationship to derive how slip varies along the member length. Finally, these models find the crack-width as the relative slip between steel and concrete along the member length.

a. Noakowski [1], [5], [6]:

This model uses the following bond-slip relation to solve the SODE:

$$\tau(x) = 0.95 R_{cm}^{0.66} s(x)^{0.12} \quad (2)$$

where R_{cm} is the mean characteristic compressive strength of concrete. Using this relation, the strain in steel can be found out as:

$$\varepsilon_s(x) = \frac{4}{E_s \phi} \int_0^x \tau(x) dx \quad (3)$$

Thus, Noakowski proposed the following crack width equation:

$$w_m = s_m \varepsilon_m = 2.33 \left[\frac{\left(\frac{k_1 f_{ctm}}{\rho} \right)^{0.88}}{R_{cm}^{0.66}} \phi \right] \left[\frac{\sigma_{s2} - 0.42 \left(\frac{k_1 f_{ctm}}{\rho} \right)}{E_s} \right] \quad (4)$$

where, k_1 is a coefficient accounting for effective stress distribution depending on the loading type (bending or direct tension).

b. Tension Chord Model [7]:

The tension chord model assumes a rigid-perfectly plastic bonds-slip behavior, i.e., $\tau_b = 2f_{ct}$, when $\sigma_s < f_y$, and $\tau_b = f_{ct}$ when $\sigma_s \geq f_y$. Using this bond-slip relation, the SODE is solved, and the crack width equation is obtained as:

$$w = \delta_s - \delta_c = s_{TCM}(\varepsilon_{sm} - \varepsilon_{cm}) \quad (5)$$

where, ε_{sm} and ε_{cm} are the mean strains in steel and concrete respectively and are found by using the equilibrium equation by considering the variation of steel and concrete strains as linear over the cracked section.

c. Modified Tension Chord Model [8]:

This model uses the bond stress relation proposed in MC 2010 [9] to solve the SODE to find the crack-width with adjusted parameters. The model was derived to find crack width in axisymmetric members; however, it can be used for non-axisymmetric members by using a correction factor. This model thus obtains the crack width as:

$$w = 2 \int (\varepsilon_s - \varepsilon_{cm}) dx \quad (6)$$

$$w = 2 \left(\frac{1}{1 + \xi} \right) [\xi \varepsilon_{s0} x_r (1 - \psi) + u_0 (1 + \psi \xi)] \quad (7)$$

where, x_r is the transfer length across which the bond stresses are transferred, ε_{s0} and u_0 are the steel strains and slip at the crack location respectively, ψ is the ratio of the mean concrete strain over the cover to the concrete strain at the interface, and ξ is a constant. This x_r is obtained in a closed form equation for the Comparatively Lightly Loaded Member (CLLM), i.e., the crack formation phase. Whereas, in the crack stabilization phase, i.e., the Comparatively Heavily Loaded Member (CHLM) case it must be obtained numerically, i.e., when the strains become incompatible.

What makes this model more sophisticated than the others is that it considers a non-linear concrete displacement profile over the cover and considers a non-local bond-slip equation that which does not further complicate the procedure.

However, the drawback of this method is that the determination of the correction factor for non-axisymmetric ties is not straightforward and requires further study.

3. Pure Shear Model [10]:

This model is different from the other analytical models since it considers a no-slip condition between the steel and concrete.

In this model, when force is applied on the reinforcement, the displacement of steel can be expressed as the following:

$$u_s(x) = u_c(x) + u_{shr}(x) + u_{Goto}(x) + u_{slip}(x) + u_{shear}(x) \quad (8)$$

where, $u_c(x)$ is the longitudinal displacement of concrete, $u_{shr}(x)$ is the shrinkage displacement of concrete, $u_{Goto}(x)$ is the displacement due to Goto cracks, $u_{slip}(x)$ is the displacement due to slip, and $u_{shear}(x)$ is the shear displacement of concrete. This model is based on the following assumptions:

- I. $u_c(x)$ and $u_{shr}(x)$ are opposite in direction, hence they cancel each other out.
- II. This model assumes an elastic behaviour of concrete, hence for simplicity $u_{Goto}(x)$ is neglected.
- III. For the ribbed bars, the slip at the interface is experimentally found out to be negligible, hence $u_{slip}(x)$ is neglected.
- IV. Shear strain is taken constant across the cover.

The crack-width equation is obtained as:

$$w_m = \frac{2\varepsilon_{si} (e^{srm\sqrt{K}} - 1)}{\sqrt{K} (e^{srm\sqrt{K}} + 1)} \quad (9)$$

$\frac{(e^{srm\sqrt{K}} - 1)}{(e^{srm\sqrt{K}} + 1)} = \varphi$ is assumed to be 1, based on the upper limit of the experimental results.

$$w_m = \frac{2\varepsilon_{si}}{\sqrt{K}} = \varepsilon_{si} \sqrt{2nd_b c (1 + \vartheta)} \quad (10)$$

RESTRAINED SHRINKAGE CRACKING MODELS:

1. Gilbert [11]:

The classical model derived by Gilbert to find the crack-width in the stabilization cracking phase assumes that the strains in steel and concrete vary parabolically over a finite distance near the crack. This distance is based on an approximation depending on the reinforcement bar diameter and the reinforcement ratio. Thus, the crack-width is obtained as:

$$w = - \left[\frac{\sigma_{c1}^*}{E_e^*} \left(s - \frac{2}{3} s_0 \right) + \epsilon_{sh}^* s \right] \quad (11)$$

where, s is the crack spacing, s_0 is the length of the transfer length on either side of the crack, σ_{c1}^* is the stress in concrete in the region beyond the transfer length. This was modified by Nejadi and Gilbert [12] to include the long-term effects of support deformation.

2. Engström [13], [14], [15] :

This model considers the cracks as non-linear springs, and uses the bond-slip relation given by Jaccoud [16], i.e.,

$$\tau_b = 0.22 f_{cm} s^{0.21} \quad (12)$$

Thus, the equation to find the mean crack-width and transfer length can be found as a function of steel stress as:

$$w = 0.42 \left(\frac{\phi \sigma_s^2}{0.22 f_{cm} E_s \left(1 + \frac{E_s A_s}{E_c A_{ceff}} \right)} \right)^{0.826} + \frac{4 \phi \sigma_s}{E_s} \quad (13)$$

$$w_{net} = 0.42 \left(\frac{\phi \sigma_s^2}{0.22 f_{cm} E_s \left(1 + \frac{E_s A_s}{E_c A_{ceff}} \right)} \right)^{0.826} \quad (14)$$

$$l_t = 0.443 \frac{\phi \sigma_s}{0.22 f_{cm} w_{net}^{0.21} \left(1 + \frac{E_s A_s}{E_c A_{ceff}} \right)} + 2 \phi \quad (15)$$

where, A_{ceff} is the effective area of concrete under tension and f_{cm} is the mean compressive strength of concrete. The second term in (13) is introduced to account for the region around the crack where the bond is lost due to internal cracks, thus, this term is not to be considered in (15) and hence not present in (15). The compatibility during the cracking process can be expressed as:

$$\epsilon_{ce} L + \epsilon_{cc} L + n w + \epsilon_c L + u = 0 \quad (16)$$

where, ϵ_{ce} , ϵ_{cc} and ϵ_c are the elastic, creep and shrinkage deformation of concrete, n is the number of cracks and u is the support movement. If the force in uncracked part of concrete is greater than the critical tensile force, i.e., $N_{cr} = f_{ctm} \left(A_{ceff} + \left(\frac{E_s}{E_c} - 1 \right) A_s \right)$, a new crack will form. Thus, iterations are performed until the tensile force is less than the critical tensile force or the maximum number of cracks are reached based on the transfer length criteria, i.e., $n_{max} = L / (2 l_t) + 1$.

3. Häßler-Combe and Hartig [17]:

This model assumes that the bond-stress is transferred on both sides of a crack over the transfer length. The mean bond-stress is considered proportional to the tensile strength of concrete and the compatibility condition used is as follows:

$$n w + \int \epsilon_c(x) dx = \Delta u \quad (17)$$

where, n is the number of cracks, w is the crack-width, $\epsilon_c(x)$ is the variation of concrete strain over the length of the member, and Δu is the support movement. This yields the crack-width as:

$$w = - \frac{1}{n} \left[L \epsilon_{ce} + \frac{1}{E_c} \int \sigma_c(x) dx - \Delta u \right] \quad (18)$$

where, ϵ_{ce} is the shrinkage strain in concrete. This model highly depends on the assumed relation of mean bond-stress and mean concrete stress.

4. Ng and Kwan [18]:

These authors proposed a rigorous analytical model for concrete cracking due to shrinkage. Unlike the previous models, this model calculates the maximum bond-slip over the transfer length and proposes a way to find how transfer lengths develop as cracking takes place until the critical transfer length value. Since the maximum bond-slip occurs at the crack, the total crack-width is mean value of maximum bond

slip multiplied by the number of transfer lengths and to find the mean crack-width we can simply divide this by the number of cracks. Thus, the equation for mean crack-width is given as:

$$w = \frac{1 + m\rho}{1 + (m\rho\psi l / \tanh(\psi l))} \left(\frac{\Delta u}{L} - \varepsilon_{cs} \right) \left(\frac{L}{n_{cr}} \right) \quad (19)$$

where, m and ρ are the modular and reinforcement ratios, ψ is a constant introduced to solve the differential equation of slip, l and L are the mean transfer length and the total length of the member respectively, Δu is the support deformation, ε_{cs} is the shrinkage strain in concrete, and n_{cr} is the total number of cracks developed.

RESULTS

Comparison of experimental crack-widths with analytical results:

For the cases of cracking in direct tension and under restrained shrinkage, a database of experimental results was collated and is compared with the crack-widths estimated by the analytical models. For the direct tension models, a database of 104 experimental results was formed from the works of various researchers [19], [20], [21], [22], [23]. For restrained shrinkage, the analytical models were compared with the classical tests performed by Gilbert and Nejadi [12]. These shrinkage tests have also been used in the respective papers of the analytical models to compare their results, and thus serves as an important benchmark to validate the formulations of these models which will further be used in the sensitivity analysis. An important assumption considered in the restrained shrinkage tests is that the yield stress of the reinforcement is assumed to be high enough such that yielding does not take place.

Figure 1 to Figure 6 show the scatter of the experimental crack-widths plotted against the theoretical values calculated using the analytical models. The dashed line represents the 45° line that indicates when the theoretical values exactly match the experimental ones. The models proposed by Leonhardt [2] and the MTCM [8] give more conservative values of crack-widths as compared to the other models which is also represented by the box and whisker plot in **Figure 8**. The shrinkage cracking models show a common outlier which is also acknowledged in the experimental results due to the extremely large crack-width recorded. This outlier is the one which can also be observed in the box plot for all models.

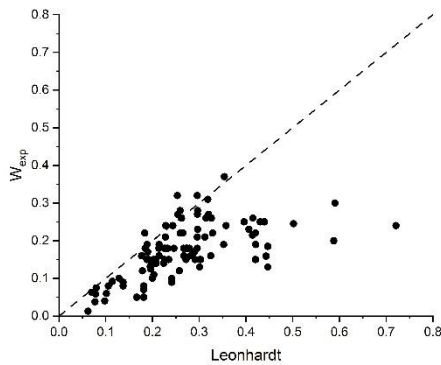


Figure 1. Comparison with Leonhardt

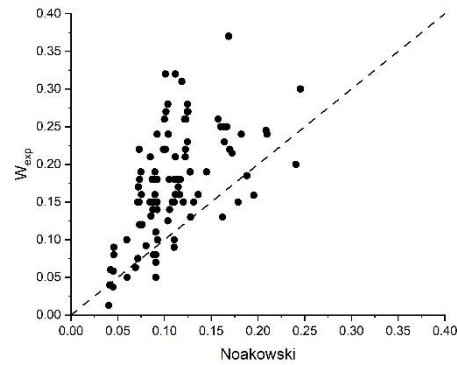


Figure 2. Comparison with Noakowski

Engström [13], [14], [15] gives highly conservative crack-widths as compared to other models which is also evident in the box and whisker plot in **Figure 7**.

Statistical parameters:

The statistical parameters are calculated for all the direct tension and restrained shrinkage cracking models and are presented in **Table 1**. For the direct tension models, the models by Leonhardt [2] and the MTCM [8] give conservative results, i.e., Mean of $W_{exp}/W_{theoretical}$ is less than 1, which is desirable for design, whereas the PSM [10] gives this value closest to 1 despite its simplicity and the assumptions considered. The MTCM [8] despite being the most robust of the direct tension models performs similar to other models when we compare the MAPE values but can be considered better since it provides conservative crack-width values with less variance and a lower RMSE value.

The restrained shrinkage models are very similar to one another except for the model by Engström [13], [14], [15] which predicts highly conservative crack-width values. The rigorous analytical model proposed by NG and Kwan [18] performs very similar to the other models and all of them have a very low MAPE

value, variance and RMSE values.

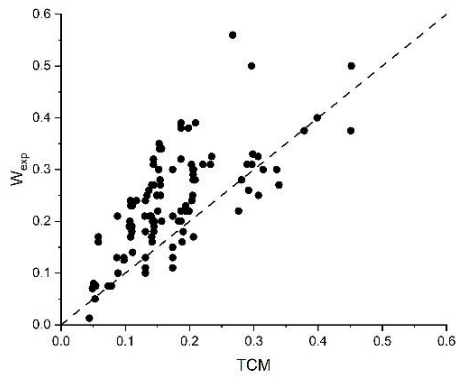


Figure 3. Comparison with TCM

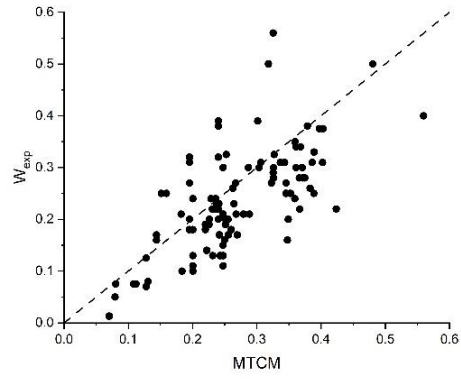


Figure 4. Comparison with MTCM

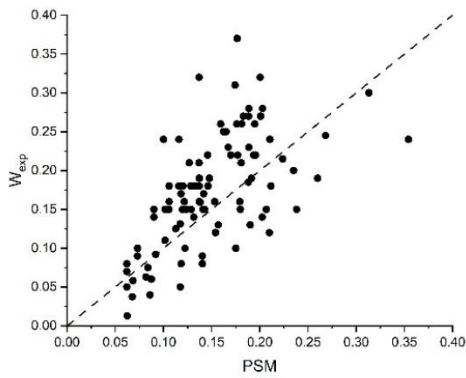


Figure 6. Comparison with PSM

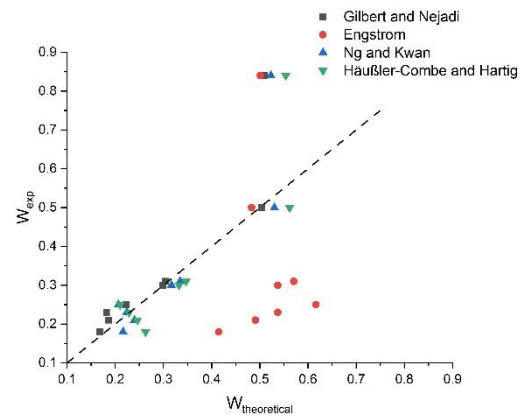


Figure 5. Comparison of Shrinkage models

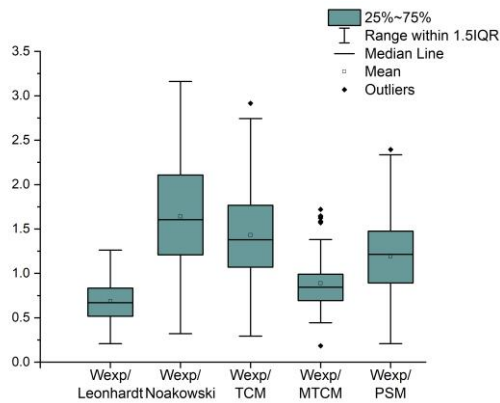


Figure 8. Box and Whisker plot of Direct tension models

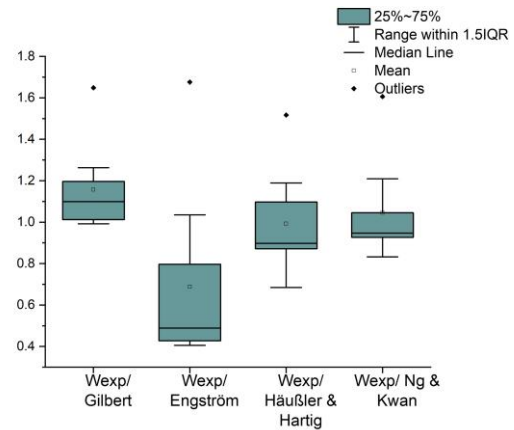


Figure 7. Box and Whisker plot of Shrinkage models

Sensitivity Analysis:

In order to investigate how different parameters affect each model and which parameter affects the most, a Sobol's [3] variance-based sensitivity analysis is performed on all the models. In this analysis, we compare the ratio of variance due to change in a particular parameter to the total variance to find the influence of that parameter. Sobol's [3] first order indices indicate the contribution of a single parameter's variance on the total variance keeping all other parameters constant, whereas the total indices also include the contribution of the correlated terms to the output. To perform this sensitivity analysis, the Latin Hypercube sampling technique was used. This is a pseudo-random sampling technique that can be understood in the following way. For n number of parameters, we have an n -dimensional space. Having N number of sampling points for each parameter implies that we divide each

parameter into N probable intervals. Thus, at a time, only 1 parameter can lie in a particular probable interval and once this parameter has appeared in a particular probable interval, it cannot repeat in the same probable interval again. This ensures a good coverage of the parameter space by using less number of sampling points. The range of the parameter values for the sensitivity analysis have been chosen from the parameter values provided in the experimental results used for the previous sections.

Table 1. Statistical results

Cracking Type	Model	Mean of $\frac{W_{exp}}{W_{theoretical}}$	Mean Absolute Percentage Error	Standard Deviation	Variance	Root Mean Square Error
Direct Tension	Leonhardt[2]	0.68	0.67	0.22	0.0496	0.5448
	Noakowski[5], [6]	1.64	0.39	0.59	0.347	1.5623
	TCM[7]	1.43	0.32	0.48	0.23	1.279
	MTCM[8]	0.88	0.36	0.29	0.086	0.6944
	PSM[10]	1.19	0.33	0.4	0.16	1.0799
Restrained Shrinkage	Gilbert and Nejadi[12]	1.16	0.12	0.14	0.0194	0.1189
	Engström[13]	0.69	0.93	0.06	0.0037	0.2744
	Häußler-Combe and Hartig[17]	0.99	0.19	0.14	0.0199	0.1106
	Ng and Kwan [18]	1.05	0.10	0.13	0.0178	0.1153

The distribution of the values is provided in **Table 2**. For restrained shrinkage, since the number of experimental results were less, the ranges have been extended keeping practical aspects in mind and presented in **Table 4**.

Table 2. Parameter range for direct tension models

Stress in steel (MPa)	Bar diameter (mm)	Mean tensile strength of concrete (MPa)	Number of bars	Elasticity modulus of Concrete (MPa)	Member Length (mm)	Sectional Area (mm ²)
100 - 450	10 - 25	2 - 4	1	22000 - 38000	750 - 5000	2500 - 202500

The ranks of how each parameter affects the crack-width in decreasing order of their influence is presented in **Table 3** for the direct tension models and **Table 5** for the restrained shrinkage models. These ranks have been obtained for 100000 model simulated outputs which show converged results. For all direct tension models, the 2 parameters that unequivocally affect the crack-width the most are the bar diameter and the sectional area of the tie. These constitute the reinforcement ratio, i.e., the ratio of area of steel to the area of concrete. An increase in the sectional area implies an increase in the concrete cover depth. Increasing the cover depth has been shown to increase the crack-spacing by virtual experiments conducted by [24] and thus it indirectly increases the crack-width. An increase in the bar diameter decreases the stress in the reinforcement and increases the equivalent axial stiffness of the member. Also, at the same stress level, a higher bar diameter specimen is exposed to a higher load which leads to more internal cracking that reduces tension stiffening and increases crack-width. Hence it has a profound effect on the crack-width. These physical phenomena help to justify the results of the sensitivity analysis. The bar diameter and the stress in steel are linked to each other, however, in the models where we deal with a fixed transfer length, the bar diameter takes precedence as it is used to empirically define the transfer length value. Thus, finding the transfer length values as a function of the stress in the reinforcement and not empirically has consequences on the crack-width and can be used as a useful point of departure for further studies.

Table 3. Parameter ranks for Direct tension models

Model	Stress in Steel	Bar Diameter	Mean Tensile Strength of concrete	Sectional Area	Member Length	Elasticity modulus of Concrete
Leonhardt [2]	3	1	4	2	-	-
Noakowski [5], [6]	4	1	3	2	-	5
TCM [7]	4	2	3	1	-	5
MTCM [8]	1	2	4	3	6	5
PSM [10]	1	3	-	2	-	4

Table 4. Parameter range for restrained shrinkage models

Area of Steel (mm ²)	Depth of section (mm)	Mean Tensile Strength of concrete (MPa)	Support Deformation (mm)	Shrinkage Strain ($\mu\epsilon$)	Elastic Modulus of Concrete (MPa)
150 - 400	90 - 120	2 - 4	0.15 - 0.45	400 - 600	22000 - 38000

Table 5. Parameter ranks for restrained shrinkage models

Model	Area of steel	Depth of Section	Mean Tensile Strength of concrete	Support Deformation	Shrinkage Strain	Elastic Modulus of Concrete
Gilbert and Nejadi [12]	1	3	2	5	4	6
Engström [13], [14], [15]	3	6	4	2	1	5
Häußler-Combe and Hartig [17]	1	3	2	5	4	6
Ng and Kwan [18]	2	4	1	6	5	3

For the restrained shrinkage models, the area of steel and the mean tensile strength of concrete play a pivotal role in determining crack-widths. The mean tensile strength is important since these models define the bond-stress in terms of the mean tensile strength and a transfer of this bond-stress from steel to concrete is what causes shrinkage cracking. The effect of the area of steel has two physical interpretations. Firstly, the increase in the area of steel is proportional to an increase in the reinforcement diameter for a constant number of bars. This increase in diameter causes an increase in the bond transfer length along the member since the transfer length is defined a function of the bar diameter. Secondly, an increase in the area of steel allows for an increase in the area of bond-stress transfer around the circumference of the bar, thus acting as an internal restraint. The magnitude of the shrinkage strain has less of an effect on the the cracking since it is not the shrinkage per se that causes cracking but the bond-stress transfer between steel and concrete.

CONCLUSIONS:

This study compares and analyzes the analytical models in direct tension and restrained shrinkage to assess the behaviour of each model. First, all the formulations are discussed, highlighting their

underlying assumptions, followed by a one-to-one comparison with the crack-widths found experimentally. Based on these results, a statistical analysis is performed on these to determine the overall behaviour of these models and finally a Sobol's [3] variance based sensitivity analysis is conducted to determine which parameters govern the most and should be explored further in order to develop a more robust codal formulation based on these analytical models.

After studying the results presented earlier, the following conclusions can be arrived at:

1. The most sophisticated direct tension model, i.e., the Modified Tension Chord Model [8], gives a conservative result for the crack-widths, with less scatter, which is desirable. Thus, the bond-stress distribution and the concrete strain profile considered should be a useful point of departure for the codal formulations.
2. Despite the realistic assumptions and complicated crack-width formulation of the MTCM [8], its MAPE value of 0.36 is not better than that of other simplistic models like TCM [7] and PSM [10] having values of 0.32 and 0.33 respectively. This implies that a simplification can be made in the iterative procedure to find the transfer length to adapt the MTCM [8] in the design codes.
3. In the restrained shrinkage cracking case, the models proposed by Gilbert and Nejadi [12], Häußler-Combe and Hartig [17], and Ng and Kwan [18], perform in a very similar manner with mean crack-width ratios of 1.16, 0.99, and 1.04 and MAPE values of 0.12, 0.19, and 0.10 respectively. Thus, adopting a mechanics-based approach of solving the second order differential equation for slip like that done by Ng and Kwan [18] and yielding a closed-form solution for crack-width will make the codal formulations more robust.
4. The difference between the older models and the new more rigorous formulation is not significant, this implies that the model by Ng and Kwan [18] can be simplified by using a fixed transfer length approach instead of finding it iteratively to implement it into the design codes.

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